

Last Time:

- Noise model
- PSD
- Allan Variance / Deviation

Today:

- Fitting a Gauss-Markov Model
- Setting up hardware

Allan Deviation Review!

- We want to characterize noise on a gyro, include bias + drift.
- Because of bias drift, standard covariance estimate doesn't converge.
- Allan deviation looks at how a moving-window average changes over time as a function of window size!

$$\bar{x}_n = \frac{1}{m} \sum_{k=m(n-1)+1}^{mn} y_k$$

moving-window avg. window index window size data

$$A(\tau) = \frac{1}{2} E[(\bar{x}_{n+1} - \bar{x}_n)^2]$$

averaging time

$$\tau = m \tau_0$$

↪ sampling time = $1/f_s$

- Allan deviation $\sigma(\tau) = \sqrt{A(\tau)}$
- Plot σ vs. τ on log scale ("σ-τ plot")

- White noise has slope $-1/2$ because longer averaging window $\rightarrow$ lower variance
- Random walk has slope $+1/2$ because longer integration time $\rightarrow$ greater "walk distance"

* MATLAB Example

- Real data doesn't perfectly follow this model:

* "Random walk" exhibits "roll off" at very long averaging times. Corresponds to "forgetting factor" on integrator.

* There is often a vertical offset / bias, at the bottom of the curve. This can be easily modeled for sim / Monte Carlo with another noise term, but it's difficult to put into the EKF (unobservable).